INCH-POUND
MIL-DTL-24643/14F
1 October 2009
SUPERSEDING
MIL-DTL-24643/14E
22 August 2002

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -20 °C TO +105 °C, 1000 VOLTS, TYPE LSSSGU

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

Construction, Watertight with Circuit Integrity

First	-	Copper conductor, coated or uncoated (see table I for size).
Second	-	Thermoset insulation (white) extruded or taped (see table I for thickness).
Third	-	An optional binder.
Fourth	-	Cross-linked polyolefin jacket (see table I for thickness.)

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TABLE I. Details.

Military part no. M24643/14	Type and size	Conductor size		Cable overall diameter max. (inch)	Insulation thickness min. avg. (inch)	Conductor resistance max. (ohms)	Jacket thickness min. avg. (inch)	Insulation resistance (megohms)	Accelerated service loading amperes
		Navy std	AWG						
-01UN	LSSSGU-50		3 (Class C)	0.520	0.035	0.209	0.040	160	240
-02UN	LSSSGU-75		1 (Class C)	0.602	0.035	0.132	0.040	135	315
-03UN	LSSSGU-100		0 (Class D)	0.669	0.040	0.105	0.050	125	375
-04UN	LSSSGU-200		0000 (Class D)	0.872	0.050	0.0525	0.050	90	580
-05UN	LSSSGU-300		300 MCM (Class D)	1.001	0.060	0.0370	0.050	80	750
-06UN	LSSSGU-400	400 (127)		1.118	0.060	0.0268	0.050	70	925
-07UN	LSSSGU-650		650 MCM (Class D)	1.371	0.060	0.0170	0.060	60	1300
-08UN	LSSSGU-800		800 MCM (Class D)	1.485	0.060	0.0138	0.060	50	1550
-09UN	LSSSGU-1000		1000 MCM (Class D)	1.620	0.060	0.0110	0.060	40	-
-10UN	LSSSGU-1600		1600 MCM (Class B)	2.010	0.080	0.00693	0.075	40	-
-11UN	LSSSGU-2000		2000 MCM (Class B)	2.210	0.080	0.00555	0.075	35	-

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REQUIREMENTS:	
Qualification required.	
INSPECTION:	
Basic Electricals:	
Conductor resistance (ohms/1000 feet at 25 °C, max.)	See table I
Voltage withstand (volts, root mean square, min.)	
Conductor to ground	5000
Insulation resistance (megohms/1000 feet, min.)	
Conductor to ground	See table I
Conductor continuity	No failure
Group A:	
Visual and dimensional	No failure
Watertightness (see MIL-DTL-24643 for limits of water leakage)	No failure
Group B:	
Cold bending cable	No failure
Thermoset proof test (percent, max.)	
Insulation (extruded insulation only)	50
Jacket (when tested at 200 °C)	50
Drip (95±1 °C)	Zero
Physicals (unaged)	
Insulation (extruded)	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	150
Jacket (cable)	
Tensile strength (lb/in^2, min.)	1300
Elongation (percent, min.)	160
Tear (lb/in thickness, min.)	35
Gas flame (1 hour)	No failure
Group C:	
Physicals (aged) air oven	
Jacket (cable)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60

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Permanence of printing (jacket) (cycles, min.)	125
Heat distortion (percent of unaged, max.)	30
Cable fill (sealant) removability	No failure
Shrinkage	No failure
Group D:	
Flame propagation (cable)	No failure
Qualification Inspection:	
Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:	
Cold working (minus 20±2 °C)	No damage
Gas flame (3 hours)	No failure
Accelerated service (Sizes 50 – 800) (see table I for load current)	No failure
Aging and compatibility (cable) (125±5 °C)	No failure
Abrasion resistance (jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Insulation	0.2
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Smoke index (max.)	
Jacket	25
Fillers	45
Insulation	45
Toxicity index (max.)	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - Type A (hardness, min.)	80

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Weathering (jacket)	No failure
Electrical moisture absorption (extruded insulations only)	No failure

UNIT ORDERING LENGTHS:

Type	Size	Feet (nominal)
LSSSGU	50 through 100	1500
LSSSGU	200 through 1000	1000
LSSSGU	1600 and 2000	500

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army – MI
Navy – SH

Preparing Activity:
Navy – SH
(Project 6145-2008-062)

Review Activities:
Army – AR, AV, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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